BUSINESS REQUIREMENTS DOCUMENT

Centralized Multi-Channel Lead Management Platform

Prepared for
Garuda Aeromedia

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Document purpose: Define the business requirements, MVP scope, implementation approach, integrations, architecture, costs, assumptions, and future roadmap for a centralized system that prevents genuine customer enquiries and follow-ups from being missed.

1. Executive Summary

Garuda Aeromedia receives customer enquiries through Instagram, WhatsApp, Meta Ads that redirect customers to WhatsApp, and direct phone calls. Because these interactions are handled separately, the business can lose visibility into potential buyers, duplicate customers across channels, and miss scheduled follow-ups.

The proposed solution is a centralized Lead Management Platform that treats every interaction first as an enquiry and converts it into a lead only when genuine purchase intent is established. Leads can then be assigned, qualified, followed up, and ultimately closed while preserving a complete communication and activity history.

The MVP deliberately keeps purchase-intent identification and phone-call capture simple and reliable: automated channel ingestion is proposed for Meta-connected channels, while phone calls can initially be entered by the employee after the call. AI-based conversation understanding is intentionally deferred to a later phase.

The primary business outcome is simple: every genuine purchase enquiry should have an owner, a status, and a next action.

2. Problem Understanding

Current state: enquiries arrive through multiple channels and are managed separately. A general enquiry should not automatically become a sales lead, while a customer expressing buying intent should be captured and followed through.

- Fragmented enquiry handling: information is distributed across Instagram, WhatsApp and phone interactions.
- Lead leakage: genuine buyers may not be recorded or may be forgotten after the first interaction.
- No common lifecycle: different team members may track status and next steps differently.
- Follow-up risk: agreed call times may remain in notes or memory rather than a reliable reminder system.
- Duplicate customers: the same person may contact the company through Instagram, WhatsApp and phone.
- Limited management visibility: managers need a single view of new, qualified, overdue and closed opportunities.

2.1 Business Distinction: Enquiry vs Lead

Customer interaction	System outcome	Reason
“What are your business timings?”	General enquiry	No explicit purchase intent.
“I want to buy Product A. What is the price?”	Create lead	Clear buying intent.
“Please call me tomorrow at 5 PM about Product A.”	Create lead + follow-up	Purchase intent plus explicit next action.

3. Business Objectives

- Centralize genuine purchase enquiries from all supported channels.
- Provide one searchable source of truth for lead information and activity.
- Ensure every active lead has an assigned owner and clear status.
- Make follow-up scheduling and reminders part of the lead lifecycle.
- Reduce duplicate leads when the same customer uses multiple channels.
- Provide management visibility through dashboard metrics and source-wise reporting.
- Keep the first version inexpensive, understandable and implementable within an internship project timeline.

4. Proposed Solution

A web-based CRM-style application will provide a unified workspace for sales staff, telecallers and management. The platform will combine manual lead capture for phone calls with automated event/message ingestion from Meta-connected channels where access and permissions are available.

Core principle: Capture → Identify → Assign → Qualify → Follow Up → Close.

4.1 MVP Feature Set

Feature	Business requirement	MVP approach
Lead Identification	Distinguish genuine purchase interest from general enquiries.	Manual/rule-based decision in V1; AI later.
Lead Creation	Store customer, source, product, enquiry, owner, status, follow-up and notes.	Lead form + API + validation.
Multi-source capture	Instagram, WhatsApp/Meta Ads and phone calls.	Meta webhooks for supported channels; manual call entry.
Central dashboard	One place to view active leads.	React dashboard with summary cards, table and filters.
Qualification	Move leads through a consistent sales lifecycle.	Status workflow and activity logging.
Assignment	Route leads to telecaller/manager/co-founder.	Role-aware assignment with audit trail.
Call tracking	Record outcome, notes and next action after a call.	Manual activity entry.
Follow-up scheduling	Store date/time against the lead.	Date/time picker + validation.
Reminders	Alert assigned user before/when follow-up is due.	Scheduled backend job + email notification.
History	Show complete lead journey.	Immutable activity/event records.
Search & filters	Find leads by source, status, owner, product and follow-up.	Server-side query filters.
Duplicate detection	Avoid multiple lead records for one customer.	Phone-number matching first; additional signals later.
Basic analytics	Show total, new, qualified, follow-ups and closed leads.	MongoDB aggregation + dashboard cards/charts.

5. Lead Lifecycle & Business Rules

5.1 Standard Lifecycle

New → Contacted → Interested → Qualified → Purchase/Closed
Alternative terminal path: New/Contacted/Interested → Not Interested.

A lead may also be reassigned or escalated. Escalation does not replace the lead status; it changes ownership or responsibility.

5.2 Core Business Rules

- Rule 1 — Enquiry is not automatically a lead: a record becomes a sales lead only when purchase intent is established by the user or future intent engine.
- Rule 2 — Active lead ownership: an active lead should have an assigned team member.
- Rule 3 — Next action: when a follow-up is required, the lead should contain a follow-up date/time.
- Rule 4 — History is append-oriented: status changes, assignments, calls and follow-ups should be recorded as activities rather than silently overwritten.
- Rule 5 — Duplicate control: phone number is the primary matching key where available; potential matches can be flagged for user confirmation.
- Rule 6 — Escalation: a telecaller can escalate a serious lead to a manager/co-founder while preserving the previous owner and activity history.

5.3 Example End-to-End Scenario

Step	System action
1	Customer sends: “I want Product A. Please share price and call tomorrow at 5 PM.”
2	Enquiry is classified as a potential purchase lead.
3	System checks phone/identity signals for an existing lead.
4	New lead is created or existing lead is updated with the new interaction.
5	Lead is assigned to a telecaller.
6	Telecaller calls the customer, records notes and sets status to Interested/Qualified.
7	Follow-up is scheduled for tomorrow 5 PM.
8	Reminder is sent before the scheduled time.
9	If serious, lead is escalated to co-founder/manager.
10	Final outcome is recorded as Purchase/Closed or Not Interested.

6. Implementation Plan

Module	Implementation approach	Key output
Authentication & roles	JWT/session-based authentication; role checks at API and UI.	Secure login and role-aware access.
Lead management	REST endpoints for create/read/update/assign/status.	Lead CRUD and lifecycle.
Meta ingestion	Verified webhook endpoints receive permitted events and normalize them into a common internal schema.	Unified channel event pipeline.
Phone-call capture	Quick-add lead form after a call; call outcome and notes stored as activities.	Fast manual call workflow.
Assignment	Lead owner stored with user ID; assignment change creates activity.	Ownership and escalation trace.
Follow-ups	Store dueAt timestamp; scheduled worker queries upcoming/overdue records.	Reliable next-action tracking.
Notifications	Transactional email service sends assignment/reminder/overdue alerts.	Email reminders.
Search/filter	Indexed fields and query parameters; server-side filtering.	Fast lead discovery.
Duplicate detection	Normalize phone numbers and check before lead creation; flag ambiguous matches.	Reduced duplicate records.
Dashboard	Aggregations calculate counts by status, source and follow-up date.	Operational visibility.

7. APIs & Third-Party Services

Service/API	Purpose	Selection reason	Cost position
Meta Graph API / Instagram messaging capabilities	Receive supported Instagram interactions.	Official ecosystem integration and webhook-based event delivery.	No separate developer subscription assumed; Meta permissions and applicable usage policies apply.
WhatsApp Business Platform / Cloud API	Receive supported WhatsApp business messages/events; support Meta Ad → WhatsApp flow.	Official WhatsApp integration path and webhook support.	API access itself is not treated as a fixed subscription; WhatsApp message charges may apply depending on current Meta pricing and message category.
MongoDB Atlas	Cloud database.	Flexible document model suits leads plus evolving activity history.	Free cluster available for development; paid Flex/dedicated tiers for production growth.
Brevo Transactional Email	Lead assignment and follow-up reminders.	Simple Node integration, transactional email support and free starter allowance.	Free plan includes 300 emails/day; paid plans available as volume grows.

Service/API	Purpose	Selection reason	Cost position
Vercel	Frontend deployment.	Fast React deployment and automatic CI/CD.	Hobby is $0 but restricted to person al/non-commercial use; business production should use an appropriate paid/business plan.
Render	Node/Express backend deployment.	Straightforward web-service deployment and Git-based deploys.	Free resources are suitable for testing, not production; paid web services start around $7/month depending on configuration.

8. Technical Architecture

Recommended stack: React.js frontend; Node.js + Express.js backend; MongoDB Atlas; Meta Graph/WhatsApp APIs and webhooks; Brevo transactional email; Vercel/Render deployment.

8.1 Logical Architecture

Customer Channels
Instagram | WhatsApp | Meta Ads → WhatsApp | Phone Call
↓
Channel Integration Layer
Meta Webhooks for supported automated channels + Manual Call Entry
↓
Node.js / Express Backend
Authentication • Lead Service • Assignment • Qualification • Follow-up Scheduler • Notification Service • Webhook Processor
↓
MongoDB Atlas
Users • Leads • Activities • Follow-ups • Notifications • Channel Events
↓
React Web Dashboard
Lead List • Lead Details • Filters • Follow-ups • Analytics • User/Assignment views

8.2 Data Flow

Flow	Description
Inbound Meta event	Meta channel → verified webhook → backend normalization → duplicate check → lead/interaction record → dashboard.
Phone call	Employee → quick-add lead → validation/duplicate check → lead + call activity → assignment → follow-up.
Follow-up	User schedules dueAt → scheduler identifies upcoming/overdue item → notification service sends email → activity/notification status recorded.
Dashboard	React UI → authenticated API → MongoDB query/aggregation → filtered lead data → user interface.

8.3 Core Data Model

Entity	Important fields
User	name, email, role, active, createdAt
Lead	name, phoneNumber, source, product, enquiry, assignedTo, status, followUpAt, notes, createdAt, updatedAt
Activity	leadId, type, description, performedBy, timestamp, metadata
Notification	leadId, userId, type, scheduledAt, sentAt, status
ChannelEvent	channel, externalEventId, externalUserId, payloadReference, receivedAt, processingStatus

8.4 Security Requirements

- Store API tokens and secrets only in server-side environment variables/secret storage.
- Verify Meta webhook authenticity and reject unverified requests.
- Use authentication and role-based authorization for dashboard operations.

- Validate and sanitize inbound webhook payloads and user input.
- Do not expose database credentials or Meta access tokens to the frontend.
- Use HTTPS in deployed environments and minimize retained customer data to what the business needs.

9. Cost Estimate

The goal is a low-cost MVP. Exact Meta/WhatsApp usage costs depend on the business account, country, message category and current Meta commercial pricing, so variable messaging charges should be treated as pass-through operating costs rather than hidden inside a fixed estimate.

For USD conversion in this document, a rounded planning rate of ₹96/USD is used; the rupee closed at ₹95.7525/USD on 19 August 2026. citeturn2news0

Item	Development / MVP	Indicative monthly production	Notes
React + Vercel	₹0	₹0–₹1,920+	Vercel Hobby is $0 but is for personal/non-commercial use; Pro is $20/month. Business production should use the appropriate plan. citeturn1search0turn1search10
Node + Render	₹0 for testing	~₹672+	Render free services are intended for testing/non-production; paid web services start around $7/month. citeturn3search0turn3search7
MongoDB Atlas	₹0	₹0–~₹2,880	Free cluster is available; Flex is $8–$30/month depending on usage. citeturn0search0turn0search7
Brevo Email	₹0	₹0 initially	Free plan provides 300 email sends/day; paid plans start at $9/month. citeturn1search2turn1search3
Meta APIs/Webhooks	₹0 fixed subscription assumed	Variable	WhatsApp/Meta commercial usage can introduce message charges; confirm against the business account and current Meta pricing before production launch.
Domain	Optional	~₹800–₹1,500/year	Depends on registrar and TLD; not required for BRD prototype.

Recommended MVP operating target

Development: approximately ₹0–₹2,000 for software infrastructure if free tiers are sufficient.
Initial production infrastructure: approximately ₹700–₹5,000/month excluding variable WhatsApp/Meta message charges and domain costs, depending on hosting/database tier and business usage.

These are planning estimates, not vendor quotations. Production selection should be finalized after expected lead volume, message volume and Meta account configuration are known.

10. Development Effort & Delivery Plan

Phase	Work	Outcome
Phase 1	Requirements, UX, database model, API contract	Approved technical blueprint
Phase 2	Authentication, users, lead CRUD, lifecycle	Working core CRM
Phase 3	Assignment, activities, call entry, filters	Operational lead workflow
Phase 4	Follow-up scheduler, reminders, email	Follow-up reliability layer
Phase 5	Meta webhook integration and normalization	Automated channel ingestion
Phase 6	Dashboard analytics, duplicate handling, security hardening	MVP-ready release

11. Future Enhancements

Enhancement	Value	Priority
AI purchase-intent classification	Automatically identify high/medium/low buying intent from messages.	High
AI entity extraction	Extract product, price request, preferred callback time and other structured fields.	High
AI lead scoring	Rank leads using intent, engagement, response history and outcome data.	Medium
Call transcription & analysis	Convert recordings to notes and detect intent/objections/follow-up commitments.	Medium
WhatsApp notifications	Send business-approved reminders/updates through WhatsApp.	Medium
Smart routing	Automatically route leads by product, geography, workload or lead score.	Medium
Sales funnel analytics	Conversion rates, source ROI, response time and team performance.	High
SLA/overdue escalation	Escalate uncontacted or overdue high-value leads automatically.	High
Customer 360	Combine interactions, purchases and future service history into a customer profile.	Long-term

12. Assumptions & Constraints

- The business will provide appropriate Meta Business, Instagram and WhatsApp Business assets and permissions for integration testing.
- Instagram/WhatsApp automation is subject to Meta API eligibility, permissions, webhook configuration and applicable platform policies.
- Phone calls are manually captured in V1; automatic call recording/transcription is not a mandatory MVP requirement.
- The first version targets a small internal sales team and modest lead volume; infrastructure can scale after actual usage is measured.
- Email is the initial notification channel to control cost and implementation complexity.
- AI is deliberately outside the critical path for V1 so the core workflow remains deterministic and testable.
- The system should retain only business-required customer information and follow applicable privacy and consent requirements.

13. Acceptance Criteria

- A user can create a lead with all required fields and identify its source.
- A lead can be assigned, reassigned and moved through the defined lifecycle.
- A user can record call notes and communication activities against a lead.
- A user can schedule a follow-up date/time and view today's/upcoming/overdue follow-ups.
- The system sends a reminder for a due follow-up through the configured email service.
- A user can search/filter by source, status, assigned person, product and follow-up date.
- The system checks for duplicate customers before creating a new lead when a phone number is available.
- Dashboard metrics reflect current lead/status/source/follow-up data.
- Supported Meta webhook events can be verified, processed and mapped into the internal lead/activity model.

- Access to customer data is protected by authentication and role-based authorization.

14. Conclusion

The proposed platform converts a fragmented, multi-channel enquiry process into a measurable sales workflow. Instead of treating every message or call as a lead, it establishes a clear distinction between general enquiries and genuine buying opportunities. Once a lead exists, ownership, status, communication history and next action become visible in one system.

The MVP focuses on solving the business problem before adding AI. This reduces cost and delivery risk while creating the structured data foundation required for later AI-powered intent classification, lead scoring and conversation intelligence.

The key success metric is not simply the number of leads stored. It is the reduction of missed genuine enquiries and missed follow-ups, supported by clear ownership and measurable lead progression.

15. Reference Notes

Vendor information used for the cost and architecture assumptions was checked against current official documentation/pricing pages on 19 August 2026: MongoDB Atlas pricing/free tier, Vercel pricing, Render deployment/pricing documentation, and Brevo transactional email limits/pricing. Meta integration costs and eligibility should be validated against the business's own Meta account before production implementation.

Official references: MongoDB Atlas □url□Pricing□https://www.mongodb.com/pricing□ • Vercel □url□Pricing□https://vercel.com/pricing□ • Render □url□Docs□https://render.com/docs/web-services□ • Brevo □url□Transactional Email□https://www.brevo.com/products/transactional-email/□